RIYAZ MOHAMMAD
SENIOR DATA ENGINEER
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PROFESSIONAL SUMMARY

- Overall **9+ years** of experience in designing, implementing, and optimizing large-scale **data engineering** solutions, with a deep understanding of data pipelines, **cloud services**, and distributed computing.
- Designed and implemented cloud-based solutions with **AWS Lambda, Azure Event Hubs** and **AWS Glue** for server less data processing and automation of **ETL** tasks.
- Solid experience in data migration projects, including migrating on-premises systems to cloud platforms (**Azure**, **AWS**), ensuring minimal downtime and data integrity.
- Proven expertise in developing and managing end-to-end data pipelines using **Azure Data Factory** (**ADF**) for data ingestion, transformation, and storage in cloud environments.
- Strong background in building and maintaining data lakes with **Azure Data Lake Storage Gen2**, handling structured, semi-structured, and unstructured data at scale.
- Experience in leveraging **Informatica Power Center** and **Informatica Cloud Data Integration** for complex **ETL** workflows to integrate data from diverse sources and optimize data flows.
- Advanced proficiency in **Python** for data processing, analysis, automation, using libraries such as **NumPy**, **SciPy, scikit-learn, and TensorFlow** for predictive analytics and machine learning.
- Expertise in building and optimizing **Spark**-based machine learning models using **MLLib**, performing real-time analytics and processing large datasets efficiently.
- Demonstrated ability to optimize **PySpark** and **Hadoop** jobs, reducing job execution times and resource consumption through techniques like partitioning, caching, and broadcast joins.
- Hands-on experience with **Kafka** for high-throughput data ingestion, designing fault-tolerant systems with replication and partitioning strategies for high availability.
- Expertise in managing and scaling **Hadoop** ecosystems, leveraging tools such as **Hive, Impala** and **Map Reduce** for distributed data storage and parallel processing.
- Proficient in managing **SQL** and **NoSQL** databases such as **PostgreSQL**, **MongoDB**, and **Cassandra**, optimizing performance, and ensuring data consistency in large-scale applications.
- Skilled in using **Tableau** and **Power BI** to develop interactive dashboards and visualizations, enhancing data accessibility and decision-making across organizations.
- Proven ability to implement data governance frameworks and security controls for both structured and unstructured data, ensuring compliance with regulatory standards.
- Extensive experience with **Airflow** for orchestrating and automating data workflows, optimizing task dependencies, and ensuring seamless data pipeline execution.
- Adept at performance tuning for **SQL** queries and data processing jobs, reducing processing times, enhancing resource usage, and improving overall system efficiency.
- Managed data warehouse solutions using **Snowflake** for scalable, cloud-based storage, and real-time analytics across diverse business departments.
- Led the implementation of **Git**-based version control for data engineering projects, ensuring streamlined collaboration, consistent code quality, and traceable project histories.

TECHNICAL SKILLS:

Programming Languages	Python SQL, NoSQL.
Cloud Platforms	Azure, AWS, Snowflake.
ETL Tools	Informatica Power Center, Informatica Cloud, Apache NiFi, AWS Glue
Big Data Tools	Hadoop Ecosystem, Spark, Airflow.
Databases	PostgreSQL SQL, NoSQL, Snowflake, Hive, Impala.
Data Visualization	Tableau, Tableau Server, Tableau Online, Power.
Workflow Automation	Apache Airflow, Apache Oozie.
Data Migration	On-Premises to Cloud Migrations (Azure, AWS), ETL Automation Scripts, Kafka, Legacy System Integrations.
Performance Optimization	SQL Query Optimization, PySpark, Hadoop Resource Tuning, Airflow Pools.
Collaboration Tools	Jira, Git.

PROFESSIONAL EXPERIENCE

Client: Stifel Financial Corp., St. Louis, MO **Feb 2023 – Present**
Role: Sr. Data Engineer
Responsibilities

- Designed and developed end-to-end data pipelines using **Azure Data Factory** (**ADF**) for ingestion, transformation, and loading of large-scale data from diverse sources.
- Managed and monitored data storage using **Azure Data Lake Storage Gen2** to handle structured, semi-structured, and unstructured data at scale.
- Designed and developed complex **ETL** mappings and workflows using **Informatica Power Center** to extract, transform, and load data efficiently.
- Implemented data quality rules using **Informatica Data Quality** (**IDQ**) to ensure accurate and consistent data across systems.
- Utilized **Python** libraries such as **NumPy** and **SciPy** to perform statistical analysis and numerical computations on datasets.
- Optimized **Python** scripts for performance by leveraging multi-threading, multiprocessing, and efficient data structures.
- Built machine-learning pipelines using **Spark MLLib** for predictive analytics and large-scale model training.
- Implemented performance-tuning techniques by optimizing shuffle operations, caching, and parallelism settings in **Spark**.
- Wrote optimized **PySpark Data Frame** operations for aggregating, joining, and transforming petabyte-scale datasets.
- Tuned **PySpark** applications by optimizing memory usage, partitioning, and configuring parallel execution for large jobs.
- Developed **Kafka** producers and consumers for data ingestion, processing, and delivery in high-throughput environments.
- Ensured fault tolerance and high availability by configuring **Kafka** clusters with replication and partitioning strategies.
- Designed and implemented complex **SQL** queries to extract, manipulate, and analyze structured data from relational databases.
- Designed and optimized data models and queries for **NoSQL** databases, leveraging key-value, document, column-family, and graph structures to ensure high performance for large datasets
- Ensured data consistency and high availability by setting up replication, clustering, and failover configurations in **PostgreSQL** environments.
- Implemented and optimized **Hadoop** ecosystems by leveraging **HDFS** for distributed data storage and Map Reduce for parallel data processing, improving data processing speed and scalability.
- Monitored and tuned the **Hadoop** clusters, ensuring optimal performance, resource allocation, and fault tolerance, thereby reducing system downtime and improving overall efficiency.
- Integrated various data sources and systems with **Airflow** by implementing connectors to **APIs**, databases, and cloud services, facilitating seamless data movement across platforms.
- Designed and implemented **Snowflake**-based data warehouses, enabling centralized storage and real-time analytics for large-scale data ingestion and reporting across multiple departments
- Developed interactive dashboards and reports in **Power BI**, enabling real-time data visualization and actionable insights for business decision-making.
- Led the implementation of **Git-based** version control for data engineering projects, ensuring seamless collaboration, maintaining a clean commit history,
- Managed and prioritized backlogs within **Jira**, ensuring that the team remained focused on the most critical tasks related to data engineering pipelines, infrastructure, and performance optimization for on-time delivery.

Environment: Azure Data Factory, Data Lake Storage Gen2, Informatica Power Center, Informatica Data Quality, Python, NumPy, SciPy, Spark MLLib, PySpark, Kafka, SQL, NoSQL, PostgreSQL, Hadoop, HDFS, Map Reduce, Airflow, Snowflake, Power BI, Power Query, Power Pivot, DAX, Git, Jira.

Client: Allstate, Northbrook, IL **Dec 2019 – Jan 2023**
Role: Sr. Data Engineer
Responsibilities

- Designed and implemented server less solutions using **AWS Lambda** for automated data ingestion, processing, and error handling.
- Developed **AWS Cloud Formation** templates and used Terraform for managing and automating **AWS** infrastructure deployments.
- Implemented **AWS Step Functions** to orchestrate complex workflows, ensuring seamless integration and coordination of multiple **AWS** services for data processing pipelines.

- Configured and managed **Amazon SNS (Simple Notification Service)** for real-time alerts and notifications, ensuring timely communication for critical events and system monitoring.
- Developed and containerized **Python** applications using **Docker** for portability across cloud and on premise environments.
- Created machine learning pipelines using **Python** libraries such as **scikit-learn** and **TensorFlow** to support predictive analytics.
- Designed fault-tolerant **Spark** jobs with check pointing, retries, and error-handling mechanisms.
- Leveraged **Spark** Streaming to process and analyze real-time streaming data with low latency.
- Managed data workflows in Databricks using **PySpark** notebooks to analyze and process datasets interactively.
- Applied **PySpark** window functions and user-defined functions (UDFs) for complex data transformations,
- Managed **Kafka** topics and partitions to optimize performance, maintain data integrity, and handle high volumes of streaming data.
- Monitored and troubleshot **Kafka** clusters, addressing performance bottlenecks and ensuring minimal downtime.
- Resolved complex **SQL** query performance issues, ensuring fast and accurate retrieval of large datasets.
- Implemented security controls and encryption in **NoSQL** databases to meet regulatory compliance and secure sensitive data.
- Implemented **PostgreSQL** partitioning strategies to improve query performance and manage large volumes of data in **OLAP** use cases
- Automated data **ETL** workflows using **Hadoop** tools such as **Apache NiFi** and **Oozie**, enabling more efficient and reliable data ingestion and transformation processes.
- Troubleshot and resolved complex data processing issues in **Hadoop** clusters, including node failures, slow query performance, and data inconsistency issues.
- Managed **Airflow's** metadata database to ensure efficient storage and retrieval of execution logs, task history, and system metrics, enabling data-driven workflow improvements.
- Managed the sharing and collaboration of **Power BI** reports and dashboards through the Power BI Service, creating workspaces, publishing reports, and setting up access controls for multiple users across the organization.
- Configured and maintained automated data refresh schedules in **Power BI** Service to ensure data accuracy and timeliness in reports, leveraging data gateways for on-premises data sources.
- Conducted code reviews using **Git** pull requests, providing constructive feedback to team members, ensuring coding standards, and optimizing data pipelines to improve performance, scalability, and maintainability.
- Led sprint planning sessions in **Jira**, defining clear user stories, estimating tasks, and assigning tickets to team members to ensure efficient execution of data-related initiatives.

<u>Environment:</u> AWS Lambda, AWS CloudFormation, Terraform, AWS Step Functions, Amazon SNS, Docker, Python, scikit-learn, TensorFlow, Spark, Spark Streaming, Databricks, PySpark, Kafka, SQL, NoSQL, PostgreSQL, Hadoop, Apache NiFi, Oozie, Airflow, Power BI, Git, Jira.

Client: Yashoda Hospitals, Hyderabad, India **Apr 2017 – Oct 2019**
Role: Data Engineer
<u>Responsibilities:</u>

- Migrated on-premises data systems to **Azure** cloud environments, ensuring minimal downtime and high performance.
- Designed and managed data pipelines in **Azure Data Factory** for automated data ingestion, transformation, and loading processes, ensuring reliable and efficient data workflows
- Leveraged **Azure Event Hubs** and **Azure Stream Analytics** to process and analyze real-time streaming data from IoT devices.
- Integrated cloud-based data sources with **Informatica Cloud Data Integration** to support hybrid cloud architectures for data movement.
- Utilized **Python's** multiprocessing capabilities to parallelize resource-intensive tasks and enhance the performance of large-scale data processing.
- Designed and executed data migration projects by writing **Python** scripts to transfer data from legacy systems to cloud-based platforms.
- Implemented **Spark SQL** for querying large datasets, optimizing performance through partitioning, indexing, and caching strategies.
- Utilized **Spark Streaming** for real-time data processing, enabling immediate insights and reporting on streaming data.
- Optimized **PySpark** code using broadcast joins and partitioning techniques to minimize shuffling and speed up execution time for massive datasets.
- Automated **Kafka** topic management using scripts and configuration management tools to streamline operations in cloud and on-premises environments.
- Led the migration of legacy message queues to **Kafka**, resulting in improved scalability, lower latency, and cost efficiency for enterprise-level applications.
- Coordinated with cross-functional teams to collect requirements and develop **SQL**-based reporting solutions for business intelligence tools.

- Managed backup and disaster recovery processes for **NoSQL** databases, ensuring data integrity and minimal downtime.
- Utilized **PostgreSQL** extensions to extend functionality and support specialized workloads like geospatial data and time-series analysis.
- Developed and implemented data models on top of **Hadoop** ecosystem tools **(Hive, Impala)** to facilitate easy querying and reporting for stakeholders.
- Developed **Tableau** visualizations such as heat maps, scatter plots, and geospatial maps to represent complex data patterns and trends, improving the clarity of business data insights.
- Enforced comprehensive documentation practices within **Git** repositories, including README files, commit messages, and tagging conventions,

Environment: Azure(ADF), Event Hubs, Stream Analytics, Informatica Cloud Data Integration, Python, multiprocessing, Apache Spark, MLLib, PySpark, Kafka, SQL, PostgreSQL, Hadoop, Hive, Impala, Airflow, Tableau, Tableau, Git.

Client: Mahindra and Mahindra Limited, Mumbai, India Jun 2015 – Mar 2017
Role: Data Analyst
Responsibilities:

- Utilized **AWS services** such as **AWS S3, EC2** and **AWS Redshift** for efficient data storage, processing, and analysis, ensuring seamless data integration across cloud platforms.
- Employed **AWS Athena** for querying structured and unstructured data stored in **AWS S3**, facilitating quick data retrieval and cost-effective analytics.
- Developed **Python** scripts to automate data processing tasks, including data cleaning, transformation, and aggregation, improving workflow efficiency.
- Integrated **APIs** using **Python** to collect and analyze data from external sources, streamlining data import and enhancing reporting capabilities.
- Utilized **Hive** and **Pig** for querying and managing large datasets in **Hadoop**, enabling high-level data analysis and reporting.
- Automated data analysis tasks using **SAS macros**, improving workflow efficiency and reducing the need for manual data manipulation.
- Integrated **SAS** with **SQL** and other data management tools for streamlined data extraction, transformation, and analysis.
- Developed **SQL** and **PL/SQL** procedures, functions, and triggers to automate data processing and ensure data integrity across systems.
- Created visually compelling dashboards using **Excel** charts and graphs, enabling stakeholders to easily interpret data and make informed decisions.
- Optimized **Tableau workbooks** by implementing advanced calculations, data blending, and data extracts to enhance performance and minimize load times for large datasets.

Environment: AWS, S3, EC2, Redshift, Athena, Python, Hive, pig, Hadoop, SAS macros, SQL, PL/SQL Excel, Tableau.

EDUCATION: BTech in Computer Science and Engineering, June 2011 - May 2015 at JNTUH, India.